

The MOND acceleration scale tracks the Hubble rate: a registered prediction and its first confrontation

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Series: Event Density (ED) — standalone data-confrontation paper **Status:** Draft for external circulation. Self-contained: no ED result is assumed that is not stated and tiered here. **Genre:** A registered prediction, its first data, and an honest verdict.

Preamble — what this paper does NOT claim

1. **It does not claim confirmation.** The prediction's *qualitative* content is confirmed and its *quantitative* rate is mildly disfavoured by the first survey to test it. Both halves are stated.
 2. **It does not claim to derive $H(z)$.** The Hubble history is inherited cosmological input throughout.
 3. **It does not rest on the $1/(2\pi)$ coefficient.** That coefficient is, in the source corpus's own current tiering, **postulated and disputed**. The claim tested here is the *exponent*, which is independent of it (§3.3). This is stated up front because it is the paper's main structural point.
 4. **It does not claim novelty for the a_0 -cosmology connection.** That connection is standard MOND material (Milgrom 2020 is a review of it). What is offered here is a *specific, falsifiable exponent* and a mechanism that cannot bend it.
 5. **It does not claim that ED is established.** This paper is one prediction from a larger substrate program (§7). It stands or falls on its own.
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Abstract

The MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ is, in MOND, a constant of nature. Its numerical proximity to cH_0 has been noted for decades and is either an accident or a structural fact. The two readings differ observationally: if the coincidence is structural and a_0 is set by the cosmic horizon $R_H = c/H$, then a_0 is **not** constant — it tracks the instantaneous Hubble rate, $a_0(z)/a_0(0) = H(z)/H_0$, growing by $\times 1.8$ at $z = 1$ and $\times 3$ at $z = 2$.

Event Density, a substrate framework, makes this reading structural and registers $a_0(z) = cH(z)/(2\pi)$ with the exponent fixed at **exactly one**. The exponent is the testable content: it cannot be tuned within the framework without abandoning the mechanism that produces it.

The first survey able to test it has reported. MUSE-DARK III (A&A 2026) detects evolution of a_0 at $\sim 30\sigma$, excluding a **constant a_0** — MOND’s classical reading of its own scale, though not the only MOND (§1.2). (*The predicted local normalisation also lands within $\sim 8\%$, but it rests on a coefficient nothing upstream fixes and is **not** offered as evidence; see §3.2 and §5.2.*) **The survey publishes a linear fit, not a power law** (§5.1). Converting it and propagating the terms the published bar omits — the intercept, the cosmology, and the reduction choice — gives **$\alpha = 1.15 \pm 0.07$, putting the prediction’s $\alpha = 1$ at 2.3σ** (§5.3). The survey team’s own analysis independently describes the evolution as *faster than $H(z)$* , so the direction of the tension is theirs even though the number is ours.

A caveat that is larger than the tension, stated here rather than in a footnote — and one that needs nobody’s cooperation to settle (§5.3c). A 2026 re-analysis of the survey’s own public catalogue (§5.3a) reports three baryonic-mass-budget systematics, each claimed sufficient to produce the trend. **Their effect sizes are quantitatively coherent with published numbers** (checked in event-density/theory/a0z_baryonic_systematics_check.py), and they bear on the $\sim 30\sigma$ detection itself, not merely the rate. **The survey’s own text confirms the first of the three:** its baryonic model carries stellar disk, atomic gas and bulge, **no molecular gas**, and the authors put ~ 0.2 dex on that omission against a **0.334 dex trend** (§5.3c). **If it holds, the honest status of this prediction is neither confirmed nor refuted but UNTESTED**, and the 2.3σ above falls with the detection it is measured against. **The catalogue is public, so this is decidable rather than merely arguable.**

The honest verdict is split: the qualitative call is confirmed and MOND’s constant scale is excluded; the specific rate is mildly disfavoured. We state the decisive test — a direct raw-data fit of α across surveys with a full error budget — and the condition under which the prediction should be abandoned.

1. Introduction

1.1 A coincidence that has to be one thing or the other

MOND fits galaxy rotation curves with a single acceleration scale a_0 . Empirically $a_0 \sim 1.2 \times 10^{-10} \text{ m s}^{-2}$, and $cH_0 \sim 6.8 \times 10^{-10} \text{ m s}^{-2}$, so $a_0 \sim cH_0/6$. The proximity has been remarked on since the earliest MOND papers and is discussed at length in the MOND literature (Milgrom 2020 and references therein). **We claim no novelty for noticing it.**

What the coincidence has never had is a *consequence*. Read as an accident, it predicts nothing. Read as structural — a_0 is set by the cosmic horizon — it predicts something sharp and immediate: **a_0 is not a constant of nature, and it should have been evolving all along.**

1.2 Four accounts, not two — and the fourth is the real rival

It is tempting to set this up as a two-horse race against MOND. **That would be wrong, and §1.1 has already said why:** the a_0 –cosmology connection is standard MOND material, so a

MOND theorist who expects a_0 to track the cosmic scale is **not refuted by evolution being found — they are vindicated by it**. The field contains at least four positions:

#	account	what it says about a_0	status after §5
1	Λ CDM	there is no fundamental a_0 ; the appearance of one is a coincidence of galaxy assembly — and that coincidence is itself expected to drift with redshift	not excluded, and not even strained (see below)
2	MOND, a_0 constant	a_0 is a constant of nature, on a par with c or G	excluded at 29σ
3	MOND with an evolving a_0	a_0 tracks the cosmic scale; the exponent is not predicted and is fitted to data	fits, by construction
4	this prediction	a_0 tracks the live horizon with the exponent fixed at exactly 1 , and no freedom to move it	alpha = 1 at 2.3σ on a computed conversion budget (§5.3), before the survey’s own systematics

Position 2 is the one that dies at high significance, and it is not the only MOND. Against **position 3** — the account that actually competes — this paper’s discriminating content is **not that a_0 evolves. It is the specific value $\alpha = 1$** . And that is precisely the number currently under tension (§5.3).

So the honest statement of what is at stake. Position 3 can absorb any exponent the data return, including the measured 1.18; it therefore fits better than position 4 does right now, and will continue to fit whatever is measured next. **Position 4 cannot move.** The distinction between them is not which fits today — it is that **only one of them made a commitment that could fail**.

That distinction is worth something only if the commitment survives. An untunable exponent is a virtue when it is right and a refutation when it is wrong; it is never a defence. **This paper is therefore a test of position 4, not an argument for it**, and §6 states the condition under which it should be abandoned.

And position 1 is not as untouched as the table suggests. Hydrodynamical simulations of plain Λ CDM with baryons — **Magneticum** (*MNRAS*, 10.1093/mnras/stac3017) — already produce an **apparent** rise in the RAR acceleration scale with redshift, from ordinary galaxy-

assembly and baryon-modelling effects, of a size comparable to what is observed. **No modified gravity, no horizon mechanism.** So:

The detection of evolution, by itself, does not discriminate this prediction from Λ CDM either. It excludes only position 2. **Positions 1, 3 and 4 all accommodate an evolving a_0 ,** and they separate only on its *rate and shape* — which is precisely the part under tension (§5.3).

What a measurement at $z \sim 1$ settles: it separates 2 from {1, 3, 4} decisively — that has now happened — and it separates 1, 3 and 4 from each other **only through the exact rate**, which is where all the remaining work is.

1.3 What this paper does

States the prediction and — precisely — how much of it is derived and how much is postulated (§3); says why the exponent is the falsifiable content and cannot be tuned (§3.3, §4); presents the first confrontation, including the part that goes against the prediction (§5); and names the decisive test and the kill condition (§6).

2. The prediction

a_0 is identified with a projection of the cosmic decoupling surface at $R_H = c/H$, giving

$$a_0(z) = \frac{c H(z)}{2\pi}, \quad \frac{a_0(z)}{a_0(0)} = \frac{H(z)}{H_0}.$$

Numerically: **$\times 1.3$ at $z = 0.5$, $\times 1.8$ at $z = 1$, $\times 3$ at $z = 2$.** With $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $a_0(0) = 1.08 \times 10^{-10} \text{ m s}^{-2}$, against a measured local $\sim 1.2 \times 10^{-10}$ — an $\sim 10\%$ match with no fitted parameter.

Downstream, the baryonic Tully–Fisher relation $v_{\text{flat}}^4 = G M_b a_0$ inherits the evolution: its normalisation is not cosmologically constant, and the high- z BTFR and radial-acceleration-relation scales move with $H(z)$.

3. What is derived, what is inherited, what is disputed

This section is the paper’s structural core. A prediction is only worth testing if it is clear which part of it is load-bearing.

3.1 Derived: the scale

That a_0 is of order cH_0 — that the MOND scale is set by the cosmic horizon rather than being an independent constant — is reached in the source corpus by **three independent routes**: the horizon-projection argument, a horizon-competition threshold argument, and a symmetric thermal-matching argument. **Three routes to the scale.**

3.2 Disputed: the coefficient

None of the three routes delivers the coefficient. The $1/(2\pi)$ is, in the corpus’s own current tiering, **postulated and disputed**: the factor cancels in the displayed algebra of the source derivation and is reinstated at an assembly step fixed by a normalisation choice that nothing upstream determines. **We flag this rather than defend it.** The corpus re-tiered it from *derived* to *postulated/disputed* in September 2026, and the claim that a_0 is “parameter-free” was withdrawn with it.

A consequence worth stating early, because it removes the paper’s most quotable number. Since K is fixed by a choice rather than by the derivation, **the agreement between $K \propto H_0$ and the measured a_0 cannot be used as evidence for K .** Any paper making this prediction has to decline its own best-sounding sentence, and this one does (§5.2).

3.3 The exponent is independent of the coefficient, and that is the point

Write the prediction as $a_0(z) = K * c * H(z)^\alpha$. The disputed quantity is K . The tested quantity is α .

- Measuring the **shape** of $a_0(z)$ tests α and is **independent of K entirely** — any constant prefactor drops out of a ratio $a_0(z)/a_0(0)$.
- Measuring the **normalisation** tests K , and is not attempted here.

The part of the prediction under dispute is not the part being tested. A reader sceptical of the 2π — and there is reason to be — can set it aside and still evaluate everything below.

3.4 Why $\alpha = 1$ cannot be tuned

Not by dimensional analysis. It would be convenient to say $a_0 = c * H * (\text{number})$ is the only dimensionally legal form, but that is false: the framework carries a second scale, l_P , so a bent power is dimensionally available. (*A summary elsewhere in the corpus states the dimensional argument; it is wrong, and the careful statement is the one given here.*)

By the mechanism plus the absence of a free scalar. $\alpha = 1$ follows from a_0 riding the *live* horizon $R_H = c/H$, one power of H per horizon. Bending the exponent requires a dimensionless number to bend it with, and the framework’s own structural result is that it produces no intrinsic dimensionless scalar. **So $\alpha = 1$ is unfudgeable from the inside: there is no knob.** That is what makes it a real bet — a fitted exponent would be worthless as a test.

3.6 Why the coefficient’s failure does not reach the exponent

The obvious objection to §3.3 is that formal independence is not evidential independence. Both K and α come from one mechanism; that mechanism demonstrably failed to deliver K ; so why trust it for α ? The answer is not “they are different symbols.” It is that **they are produced by different steps, and every failure is on one side of the line.**

The derivation has a **radial** step and an **angular** step.

The radial step places the cosmic decoupling surface. Substrate influence propagates at c ; the Hubble flow recedes at H ; the crossover fixes the boundary at

$$R_H = c/H.$$

That is a competition between two rates. **It contains no angular integral, no mode decomposition, and no normalisation choice.** In the source audit it is a D-via-I step with H inherited.

The angular step projects that surface’s leading anisotropic mode onto the accelerating chain’s residual $S0(2)$ symmetry — the $|m| = 1$ azimuthal-Fourier projection of a dipole — and this is where $1/(2\pi)$ was supposed to come from.

Now locate the failures. In the source paper’s own audit table:

step	audit tier	which part
decoupling surface at $R_H = c/H$	D-via-I	radial
the dipole is the leading anisotropic mode	D	angular, and sound
cosmic content seen as anisotropic at first order	A — analogy	angular
$\rho_0 = cH_0$ “after dimensional bookkeeping”	P — normalisation choice	angular
the surviving azimuthal factor $1/(2\pi)$	DISPUTED (<i>was D</i>)	angular
$N_{\text{substrate}} G_{\phi}^{-1}$ set “to the value that returns the result”	P — nothing upstream fixes it	angular

Every disputed, postulated and analogical row is in the angular step. The radial step carries none of them.

And the exponent is entirely radial. a_0 is an acceleration set by the horizon radius, $a_0 \sim c^2/R_H$; with $R_H = c/H$ that is $a_0 \sim H^2$. **One power of H , because there is one horizon.** The evolution $a_0(z) = K c H(z)$ follows from the same surface being read *at the epoch in question* rather than frozen at some earlier one. **Neither statement passes through the angular machinery.**

A second, independent check. The scale $a_0 \sim cH_0$ is reached by **three routes** — the dipole projection, a horizon-competition threshold argument, and a symmetric thermal-matching argument. **Two of the three do not use the dipole projection at all.** If the H -dependence were an artifact of the angular step, those two routes could not arrive at the same scale. **They do.** So the dependence on H survives deleting the entire machinery that failed.

The honest residual, stated because it is what the argument actually rests on. Two things carry the exponent, and only the first is multiply-derived:

1. $R_H = c/H$ — the radial crossover, reached three ways.

2. **The horizon is read LIVE, not frozen** — a_0 tracks the contemporaneous R_H rather than a relic value.

Point 2 is the mechanism's core physical assertion, not a theorem. It is what makes the prediction a prediction: a frozen horizon would give $\alpha = 0$ and would be indistinguishable from MOND. **It is not derived from the two alternative routes, which establish the scale rather than its time-dependence.** A reader who rejects the live-horizon reading rejects the prediction — and that is the correct place for the argument to be attackable, because it is the place where the physics is.

So the answer to the objection is: the coefficient failed in the angular step, the exponent lives in the radial step, and the radial step has three independent supports. The scepticism the 2π deserves does not transfer — but the live-horizon assumption is a separate exposure, and it is named here rather than buried.

3.5 Inherited

$H(z)$ itself; H_0 ; the MOND interpolating function; all numerical values.

4. What would distinguish this from a fit

A prediction registered after the data is worth little. Three things are stated for the record:

1. **The exponent was fixed at 1 before the data existed**, and is registered as such in the source corpus's prediction list with the $\times 1.8$ -at- $z=1$ figure explicit.
 2. **The framework cannot accommodate $\alpha \neq 1$** without abandoning the horizon mechanism (§3.4).
 3. **The prediction is falsifiable in both directions** — $\alpha = 0$ would have killed it as decisively as $\alpha = 2$ would.
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5. The confrontation

5.1 What was measured

Ciocan et al., MUSE-DARK III (arXiv:2604.22613; A&A 2026) measure the radial-acceleration-relation scale in **79 star-forming galaxies at $0.33 < z < 1.44$** in the MUSE Hubble Ultra Deep Field.

What the survey publishes — and it matters that this is stated separately from what we infer from it:

quantity	result	whose
evolution of a_0 detected	$\sim 30\sigma$	theirs
per-bin scale	rises $\sim 1.99 \rightarrow 2.71 \times 10^{-10}$ m s ⁻² across the range	theirs
a_0 at $z \sim 1$	$2.38 \pm 0.1 \times 10^{-10}$ m s ⁻²	theirs

quantity	result	whose
parametrisation	LINEAR: $a_0(z) = a_0(0) + a_1 z$, with $a_1 = 1.59 \pm 0.1 \times 10^{-10}$	theirs
power-law index α in $a_0 \sim H(z)^\alpha$	not published by the survey	—
implied $\alpha = 1.18 \pm 0.04$ (stat)	converted by us from their linear fit	OURS

The survey fits a straight line in z . It does not fit a power of $H(z)$. The $\alpha = 1.18$ on which this paper’s central tension rests is **our conversion of their published linear parametrisation**, not a number they report. **An earlier draft of this paper mislabelled it as the survey’s measurement; that was an error and it is corrected here.**

What independently corroborates the direction is that the MUSE-DARK team themselves compare $a_0(z)$ against $H(z)$ and describe the evolution as **faster than $H(z)$** . So the sign of the tension is theirs even though the number is ours.

5.2 The half that confirms

a_0 evolves. The qualitative content of the prediction — that the MOND scale is not a constant of nature — is confirmed at high significance, and **the constant- a_0 reading is excluded at 29σ** . That is a structural claim about a 40-year-old framework, made in advance, and the data agree with it. **But it does not separate this prediction from position 3 of §1.2**, which expects evolution too; the separation from that rival rests entirely on §5.3.

The local normalisation is NOT offered here as evidence, and the reason is stated rather than buried. $K \propto H_0$ lands within $\sim 8\%$ of the measured a_0 . But by §3.2 the coefficient K is not fixed by anything upstream — the source derivation’s own audit records the assembly constant as “*set to the value that returns the result*”. **A coefficient chosen to produce a number cannot then be evidenced by that number.** To be precise about what the record does and does not support: the quantity targeted was the $1/(2\pi)$ form, not the observed a_0 , so this is not reverse-engineering from the data. **But that distinction rescues the intent, not the inference.** The most the $\sim 8\%$ establishes is that the chosen coefficient is **not absurd** — a weak consistency statement, and one that would read the same way for a range of nearby choices. **It is excluded from the evidence for this paper’s claim.**

5.3 The half that does not — with the refit shown

Taken at face value, $\alpha = 1.18 \pm 0.04$ places $\alpha = 1$ at $+4.5\sigma$ and the **prediction is excluded**. An earlier draft answered this by asserting the error bar was too small and the real tension “ $\sim 1-2\sigma$,” without producing a budget. **That was special pleading, and a referee would have been right to reject it.** The budget is now computed (event-density/theory/a0z_powerlaw_refit.py), and it does not say what the earlier draft guessed.

First, what the ± 0.04 is. Propagating only the survey’s $a_1 = 1.59 \pm 0.10$ through the

endpoint conversion reproduces $\alpha = 1.178 \pm 0.034$. **That is the published bar, and it identifies the reduction our own addendum used.** It contains their statistical error through one conversion choice and nothing else.

Second, the terms it omits, switched on one at a time.

term added	α	σ	$\alpha = 1$ at
their a_1 only (± 0.10)	1.177	0.034	5.2σ
+ the intercept (± 0.10 , assumed)	1.180	0.064	2.8σ
+ cosmology ($\Omega_m \pm 0.007$)	1.180	0.065	2.8σ
+ reduction choice	1.152	0.065	2.3σ

$\alpha = 1.15 \pm 0.07$, putting $\alpha = 1$ at 2.3σ — and 73% of that variance is missing from the published ± 0.04 .

Third, and this is the term nobody's error bar contains: a straight line in z is not a power law in $H(z)$, so their fit has no single α to disagree with. The local slope $d \ln a_0 / d \ln H$ of their own fitted line runs

z	0.33	0.50	0.75	1.00	1.25	1.44
local α	1.78	1.46	1.19	1.04	0.95	0.90

a factor of two across the range, crossing $\alpha = 1$ at $z \approx 1.1$ — inside the data. Any single quoted α is therefore the output of a reduction choice. Four defensible choices give 1.118, 1.147, 1.162 and **1.178** — and the last, the one that maximises α , is the one the corpus quoted.

The honest statement, and it is narrower than either the earlier draft's or the face-value reading: the tension is **2.3σ on conversion error alone**, before any of the survey's astrophysical systematics, which are theirs to quantify and would push it lower. **The intercept assumption is the largest single lever and the input we are least entitled to** — it is published to two significant figures with no error bar. At ± 0.05 the tension is 3.2σ ; at ± 0.15 , 1.8σ . **A referee could reasonably pick any of those rows, and the paper does not get to choose for them.**

What does not change. The direction is still corroborated by the survey team's own analysis, which compares $a_0(z)$ to $H(z)$ and describes the evolution as **faster than $H(z)$** . **The tension is real and it is against the prediction.** It is smaller and much softer than 4.5σ , and it is not a refutation. It is also not a result.

5.3b The shape comparison, which avoids the conversion entirely

Comparing one derived α to another inherits the reduction choice, so it is worth asking the question without forming an exponent at all. With the amplitude free —

which it must be, the coefficient being disputed (§3.2) — how far apart are the two curves across the measured range?

z	0.33	0.50	0.75	1.00	1.25	1.44
their fit	1.53	1.80	2.19	2.59	2.99	3.29
ED, $\alpha = 1$	1.62	1.80	2.10	2.44	2.81	3.12
deviation	+6.5%	+0.2%	-4.5%	-6.0%	-5.9%	-5.2%

They agree to 4–7% everywhere in the measured range. They are close, and they are not the same. Two things must be said against the prediction here. **The worst deviation, 6.5%, is larger than the survey’s ~4.2% precision on a single point**, and with 79 galaxies the *trend* is pinned far better than any single point, so the relevant comparison is tighter still. **And the residual is structured rather than random** — ED runs above their line at the bottom of the range and below it across the top, crossing near $z \sim 0.5$. **That is a curvature difference, which no amplitude change can absorb.** So the disputed coefficient does not rescue the shape, and more data of the same kind will sharpen this rather than wash it out.

What this does establish is what the $\sim 30\sigma$ is and is not. It is a detection that a_0 evolves. **It is not a measurement of the exponent**, and the two must not be reported as though the first settled the second.

5.3a The systematics challenge — located, read, and checked

An earlier draft of this section recorded this paper as possibly non-existent. Three searches had returned no locatable record and the one detailed description came back only after its title was supplied in a query, so it was written up as an artifact risk rather than a source. **That was wrong, and the correction matters more than the error.**

It exists. Mikheil Rusishvili (2026), “*Three baryonic systematics, each sufficient: on the reported redshift evolution of the MOND acceleration scale*,” self-posted at academia.edu/170831186. **It is not in arXiv or A&A**, which is why the searches failed, and **it is not peer-reviewed** — an independent researcher on a platform with no quality filter. That is a real caveat and it is not a reason to ignore it: the content is specific, quantitative, and drawn from **Ciocan et al.’s own public catalogue**.

Its three claims. Each is asserted to be independently sufficient to account for the trend.

#	systematic	claimed effect
1	unmodelled molecular gas — the survey models stellar disk + atomic gas only; adding Tacconi et al. (2018) fractions per galaxy (median M_{H_2}/M_* rising 0.51 $\rightarrow$ 0.97 across the four redshift quartiles)	removes 82% of the binned trend, 77% of the fractional evolution

#	systematic	claimed effect
2	dynamical-vs-photometric stellar-mass drift — dynamical M^* falls relative to photometric by 0.39 dex (3.6σ) between extreme bins, opposite in sign to expected stellar-to-halo-mass evolution	matches the +0.2 to +0.45 dex Ciocan et al. themselves state would remove the trend entirely
3	disk-halo degeneracy — median 1.04 dex across their six released halo models <i>for the same galaxy</i>	—

Its conclusion: **a_1 should not presently be used to constrain modified-gravity models.**

What we can check, and what we cannot. We do not have the catalogue, so **this paper cannot verify the analysis and does not claim to.** What is checkable from published numbers alone is whether the claimed effect sizes are **mutually coherent** — the same test the **alpha** conversion got. A critique whose arithmetic does not close can be set aside; one whose arithmetic does close cannot.

The mechanism is simple enough to state in one line, which is part of why it is serious. In the deep-MOND regime $g_{\text{obs}} = \sqrt{g_{\text{bar}} a_0}$, so the *inferred* scale is $a_0 = g_{\text{obs}}^2/g_{\text{bar}}$. **If the baryonic budget is underestimated by a factor f , the inferred a_0 is inflated by exactly f .** And if the missing fraction **grows with redshift** — which molecular gas demonstrably does — then **a_0 appears to evolve even when it is exactly constant.**

The arithmetic closes. Writing $f = 1 + (M_{\text{H}_2}/M^*)/(1 + M_{\text{HI}}/M^*)$ and scanning the unmeasurable atomic fraction:

M_{HI}/M^*	0.0	0.1	0.2	0.3	0.5	1.0
binned trend removed	86%	81%	77%	73%	67%	55%

The claimed 82% is reproduced at $M_{\text{HI}}/M^* \approx 0.1$ — and massive star-forming galaxies at $z \sim 1$ are molecular-dominated, so that is **the expected regime rather than a tuned one.** The number falls out of two published quantities, their binned trend and Tacconi’s fractions, **with no free parameter.**

Claim 2 is close to self-certifying, because the size required was stated by the survey team: the claimed 0.39 dex sits inside their own 0.2–0.45 dex window and exceeds the 0.334 dex of the fitted trend. **Claim 3 is the most damaging if true** — a per-galaxy modelling spread of 1.04 dex is **$3.1\times$ the 0.334 dex signal**, which would not bias the trend so much as deny that it is measured.

Verdict, stated against interest. All three effect sizes are quantitatively coherent, and this is worse for the prediction than the exponent tension was.

- **It attacks the detection, not the rate.** The $\sim 30\sigma$ was the one thing here being banked as a win.
- **ED loses that win, and the exclusion it was scored against.** If the evolution is a baryon-budget artifact, then constant- a_0 MOND is not excluded either, and §1.2's position 2 comes back.
- **The 2.3σ falls with it.** A tension computed against an unreliable detection is itself unreliable. **The confirmation and the tension go together; neither can be banked separately.**
- **The resulting state is not “refuted” and not “vindicated.” It is UNTESTED** — the one survey that appeared to test this prediction may not have. That returns it to open, awaiting Rubin/Euclid.

5.3c Claim 1 is confirmed at the source, and the catalogue is public

The first systematic does not need to be inferred from coherence. The survey states it. Their baryonic acceleration is built as

$$a_{\text{bar}}(r) = \frac{v_{\text{disk}}(r)^2}{r} + \frac{v_{\text{HI}}(r)^2}{r} \left(+ \frac{v_{\text{bulge}}(r)^2}{r} \right)$$

— **stellar disk, atomic gas, bulge. There is no molecular term.** And the authors name the consequence themselves: “*given typical molecular gas fractions of $\sim 30\text{--}50\%$ at $z \sim 1$, this would introduce a systematic uncertainty of ~ 0.2 dex in the total disk mass.*”

0.2 dex is their own figure, and the trend across their fitted range is 0.334 dex. The survey therefore acknowledges a systematic **60% the size of its own signal**, in the one component whose fraction is known to grow with redshift. **What the re-analysis adds is not the existence of the systematic — that is conceded — but the claim that it is *differential* in z and therefore mimics the trend rather than merely offsetting it.**

And the data is public. “*All catalogues and data products from our disk–halo decomposition, including the RCs, can be found on the DARK website*” (dark.univ-lyon1.fr/data-releases/; the MUSE UDF sample of 126 galaxies covers this paper). **So the decisive test named in §6 is executable rather than requestable.** Nothing here waits on anyone’s cooperation: the rotation curves, the disk–halo decompositions and the baryonic components are downloadable, and both open questions — the direct $a_0 \sim H(z)^\alpha$ fit and the molecular-gas correction — can be settled from them.

This paper does not claim to have run that test. It states that the test is available, that the prediction’s status is **untested** until someone runs it, and that **running it is the only thing that would change the verdict in either direction.**

5.4 The verdict, stated plainly

Qualitative call: confirmed, and MOND’s constant a_0 excluded at 29σ .
Quantitative rate: mildly disfavoured at 2.3σ on a computed conversion

budget (§5.3), from one survey.

And a caveat that outranks it (§5.3a): a re-analysis of that survey’s own catalogue reports three baryonic systematics whose effect sizes check out arithmetically and which bear on the $\sim 30\sigma$ DETECTION. If they hold, both the qualitative win and the quantitative tension go away together, and the correct verdict on this prediction is UNTESTED rather than either. This is an open tension, not a refutation — and it is not a confirmation either.

The prediction is closer to the data than MOND is by a wide margin and is not sitting comfortably.

6. The decisive test, and the kill condition

The decisive test is a direct fit of α to raw data across multiple surveys with a full error budget — MUSE-DARK galaxies plus independent $z \sim 0.9$ – 2.4 rotation-curve samples — rather than a re-reading of one published linear fit. Rubin- and Euclid-era data should settle it.

Kill condition, stated in advance:

If a multi-survey fit with a complete error budget returns α inconsistent with 1 at $\geq 3\sigma$, this prediction is refuted. There is no version of the mechanism that survives with a bent exponent (§3.4).

A weaker outcome — α consistent with 1 at 1σ — would be a genuine confirmation of a registered, unfudgeable, framework-distinctive number.

7. Provenance

This prediction comes from Event Density, a substrate framework in which the arrow of time is the single process primitive. **That framework is not argued here and nothing above depends on accepting it.** The relevant chain is Paper_029 (the horizon projection giving $\mathbf{a}_0$), Paper_037 (continuum invariance of $\mathbf{a}_0$), Paper_031 (the BTFR inheritance), and Paper_038 CO-3 (the registered forward prediction). The corpus, its claim register with per-claim tiers, and the audit trail behind the September 2026 re-tiering of the 2pi are public.

The prediction stands or falls on §5 and §6 alone.

8. Load-bearing step audit

#	step	tier	note
1	a₀ is set by the cosmic horizon $R_H = c/H$	D	three independent routes (§3.1)
2	the $1/(2\pi)$ coefficient	P — postulated/disputed	§3.2; not tested here
3	alpha = 1 from one power of H per live horizon	D	§3.4, and independent of step 2
4	no intrinsic dimensionless scalar to bend alpha	D (framework structural result)	what makes alpha unfudgeable
5	$H(z)$, H_0 , the interpolating function, all values	I	inherited
6	BTFR normalisation inherits the evolution	D-via-I	composition of step 1 with the standard BTFR
7	evolution detected at $\sim 30\sigma$; alpha = 0 excluded	I — measurement	MUSE-DARK III
8	the survey's linear fit $a_1 = 1.59 \pm 0.1 \times 10^{-10}$	I — measurement	Ciocan et al.
8b	alpha = 1.15 ± 0.07 converted from row 8 by us; alpha = 1 then at 2.3σ	D-via-I — OUR REANALYSIS, not the survey's result	corrected 2026-09-07 ; an earlier draft tagged this I -- same source, which was wrong. Refit and full budget now shown (§5.3) and reproducible: event-density/theory/a0z_powerla

#	step	tier	note
9	the published ± 0.04 omits the intercept, the cosmology and the reduction choice; a full conversion budget gives $\alpha = 1.15 \pm 0.07$, i.e. $\alpha = 1$ at 2.3σ	D-via-I (<i>was A -- position</i>)	computed 2026-09-07, not asserted: event-density/theory/a0z_powerla The residual judgement is the intercept's uncertainty, published to 2 s.f. with no error bar — at ± 0.05 the tension is 3.2σ , at ± 0.15 it is 1.8σ , and the paper does not get to pick composite
10	verdict: qualitative confirmed, rate disfavoured	A $\rightarrow$ position	

Step 9 was the paper's soft joint and is no longer one. An earlier draft asserted the tension softened to “ $\sim 1-2\sigma$ ” without producing a budget, which the adversarial review correctly called special pleading. **The budget is now computed and it did not return the convenient answer:** 2.3σ , not $1-2\sigma$. **Step 10 remains a position.** And the underlying exposure is unchanged: if the published ± 0.04 is taken as complete, the prediction is excluded at 4.5σ and §5.4 should read differently.

9. Falsifiers

- **F1 (primary).** A multi-survey direct fit returning α inconsistent with 1 at $\geq 3\sigma$ refutes the prediction.
- **F2.** A demonstration that a_0 is constant across redshift refutes it — and would also refute the $\sim 30\sigma$ MUSE-DARK III result, so this is now a claim about that survey's reliability.
- **F3.** A derivation, within the framework, of a mechanism admitting $\alpha \neq 1$ would remove the bet's content by making it fittable. **This is a falsifier of the claim's status, not of its value**, and is stated because a theory that can absorb any exponent has not predicted one.